

STAT 626, Summer 2025: Homeworks

HW# 2: Due Friday June 13, 10:00 am CST,

1. Do Problems 2.2-2.6 and 2.9 from the textbook.

2. Let X_1, X_2 be two random variables with standard deviations 1 and 2, respectively,

and covariance equal to -2.

(a) Find $\text{Var}(X_1 - 5X_2)$, $\text{Var}(X_1 + 5X_2)$ and $\text{Var}(X_1 + X_2)$.

(b) Find the constant c so that $\text{Var}(X_1 - cX_2)$ is minimized.

3. Simulate 250 values using the model in (1.2) in page 10 of the textbook, but replace the coefficients (1.5, 0.75) by the first and last digits of your TAMU UIN.

(a) Plot the two time series together using different colors. Are the two time series plots similar? Choose one: YES NO.

(b) Further compare the plot of your time series with that in Figure 1.9. Describe the similarities and differences of the two time series plots using at least three terminologies/concepts you have learned thus far in the course. Make sure you underline these concepts in your writing.

4. (Bonus) Let $w_t \sim i.i.d. N(0, \sigma_w^2)$, compute the mean and autocovariance functions of

$$x_t = w_{t-1}w_{t-2}(w_t + t),$$

and decide if $\{x_t\}$ is stationary.

HW#1: Due Friday, June 6, 10:00 am CST.

I. Do Problems 1.1, 1.3, and 1.4 from the textbook.

II. In the simple regression model $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$, one usually deals with pairs of observations $(x_i, y_i), i = 1, \dots, n$. Show that

1. $\sum_{i=1}^n (x_i - \bar{x}) = 0$, and $\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \sum_{i=1}^n (x_i - \bar{x})y_i$.
2. $S_x^{-2} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \sum_{i=1}^n c_i y_i$,

where $c_i = \frac{x_i - \bar{x}}{S_x^2}$, $S_x^2 = \sum_{i=1}^n (x_i - \bar{x})^2$.

NOTES: (i) The bonus problems tend to get deeper into some statistical concepts, they will encourage you to review/study the relevant sections of this or your earlier textbook/notes more systematically.

(ii) To get the bonus points for a problem, a complete solution must be given. No partial credits for the Bonus problems.

(iii) **Bonus HW points will be banked**, and used only in borderline cases when assigning letter grades at the end of the summer.

3. (Bonus) For the simple linear regression model, $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$,

(a) Derive the equations for minimizing the residual sum of squares (RSS):

$$Q(\beta_0, \beta_1) = \sum (y_i - \beta_0 - \beta_1 x_i)^2,$$

and give the formulae for the least squares estimates (LSE) $\hat{\beta}_0, \hat{\beta}_1$ of the regression coefficients.

(b) Show that $\hat{\beta}_1 = \sum_{i=1}^n c_i y_i$, with c_i 's as above. If $\varepsilon_i \sim N(0, \sigma^2)$, what is the distribution of $\hat{\beta}_1$?

(c) Let $e_i = y_i - \hat{y}_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i$, $i = 1, 2, \dots, n$, be the (estimated regression) residuals. Show that

$$\sum_{i=1}^n e_i = 0, \text{ and } \sum_{i=1}^n e_i x_i = 0.$$

What is the (geometrical) interpretation of the above identities?